

Yamini Karthik

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EDUCATION

San Jose State University - B.S. in Computer Science | Expected Dec 2027

- o GPA: 3.89/4.00, [President's Scholar](#), [Dean's Scholar](#)

EXPERIENCE

Break Through Tech - AI/ML Fellow

May 2025 - Apr 2026

- Selected from **3,500+ applicants** for a competitive 12-month AI fellowship; completed Machine Learning Foundations coursework as well as applied machine learning projects through AI Studio and the Women in Data Science Datathon.

Latitude AI - AI Studio Fellow

Aug 2025 - Dec 2025

- Benchmarked **instance segmentation models** using **100,000+ annotated driving images** for autonomous driving.
- Built **data processing and evaluation workflows** comparing **Mask R-CNN** and foundation models, identifying trade-offs in segmentation quality and deployment performance; presented insights and deployment considerations to stakeholders.

Cadence Design Systems - Software Engineering Sprintern

Jul 2025 - Aug 2025

- Designed & implemented a scalable **HTML-to-PDF pipeline** in **Python** using **Playwright** & **PyMuPDF** across **100+ HTMLs**.
- Automated PDF generation with dynamic **TOCs, hierarchical bookmarks, & hyperlink preservation**, reducing production time by **80%**; presented technical implementation and optimization results to engineers at **Cadence Design Systems**.

PROJECTS

[AI Travel Itinerary Planner](#) - DevNetwork {API + Cloud + AI} Hackathon 2026 - Hackathon Project

Aug 2026 - Sep 2026

- Engineered an **LLM-powered travel planning pipeline** using **Gemini**, validating free-text preferences and generating structured day-by-day itineraries from user constraints including budget, dates, trip type, and dietary preferences.
- Built **real-time travel data retrieval pipelines** with **SerpApi**, integrating flight, hotel, attraction, and restaurant data and applying preference-aware filtering before passing grounded results to the LLM.
- Designed **structured JSON generation and validation workflows** with **Pydantic**, combining retrieved data and user constraints to produce realistic, constraint-aware itineraries grounded in retrieved businesses, prices, and ratings.

[Agentic Skincare & Haircare Advisor](#) - Agentic AI Innovation Challenge 2026 - Hackathon Submission

May 2026 - Aug 2026

- Architected a **tool-calling multi-agent application using Gemini API**, with specialized agents for user intake and recommendations, integrating tools for product retrieval, ingredient analysis, profile updates, and routine generation.
- Designed an **agentic workflow that extracts user preferences from natural language**, maintains context throughout a session, and uses the resulting profile to personalize skincare and haircare product recommendations and routines.
- Built and deployed a full-stack **React + FastAPI application** with REST APIs, combining LLM reasoning, tool calling, product retrieval, ingredient compatibility analysis, and rule-based filtering.

[WiDS Datathon 2026 - Wildfire Impact Prediction \(Kaggle\)](#)

Jan 2026 - May 2026

- Developed **XGBoost prediction pipelines** to estimate wildfire evacuation-zone impacts from environmental and geographic features.
- Engineered features and applied **preprocessing, cross-validation, and probability calibration** to improve generalization.
- Increased Kaggle leaderboard score from **0.953 to 0.957** using ROC-AUC, Log Loss, and Brier Score.
- Led project execution through milestone planning, technical decisions, and team coordination.

ACTIVITIES AND LEADERSHIP

Responsible Computing Club - Mozilla Workshop Ambassador/Mozilla Industry Ambassador

Sep 2024 - May 2026

- Organized industry networking events and **10+ workshops** on AI ethics, facial recognition, and emerging technologies.

CERTIFICATIONS

- [Machine Learning Foundations](#) - Cornell Tech, 2025
- [Building with the Claude API](#) - Anthropic, 2026
- [Foundations of AI Engineering](#) - Codepath, 2026

SKILLS

Programming: Java, Python, JavaScript, HTML/CSS, C, SQL

AI/ML : Gemini API, Claude API, PyTorch, Scikit-Learn, Pandas, NumPy

Frameworks/Tools: React, FastAPI, Git, SQLite, Jupyter Notebooks, Matplotlib, IntelliJ, VS Code, Figma